From Causal Auditing to Value Maximization for Core Distribution Assets —Part III: Transformer Management

Yi Zeng

***Abstract*—Power transformers are the most critical and capital-intensive assets in power distribution networks. Conventional equipment condition monitoring heavily relies on static thresholds and invasive sensors, failing to capture the dynamic Remaining Useful Life (RUL) and economic limits of the assets. This paper proposes the third part of the Bianque System, introducing a non-invasive "Algorithmic Alchemy" framework capable of extracting dynamic survival features from existing conventional SCADA, DGA, and underlying Digital Fault Recorder (DFR) data. We propose a "Dual-Track Survival" architecture: for oil-immersed transformers, we introduce the bionic telomere degradation hypothesis, utilizing Partial Integro-Differential Equations (PIDE) with constrained boundaries and Bellman's optimal stopping theory to deeply evaluate the economic residual value of assets under the "Hayflick Limit"; for dry-type transformers, we construct a physical degradation triad (THD $\rightarrow \Delta T$ $\rightarrow$ PD) model to quantify time-varying risks before approaching the "Grade 9" bottom line. To combat dual uncertainties, the system establishes a "Six-Mirror Racing" model evaluation and a Bayesian nested Monte Carlo convergence mechanism. Furthermore, this paper establishes an ecological closed-loop via the "Laplace Platform," relying on underlying high-speed analog-to-digital conversion mechanisms to extract microsecond-level transient sudden mutation features. These features are fed back as new covariates for diagnostic ecological evolution, achieving continuous cyclical enhancement of predictive capabilities.**

***Index Terms*—Power transformers, survival analysis, predictive maintenance, optimal stopping theory, bionics, Bayesian updating.**

I. Introduction

> *"All things come into being, and I see thereby their return."*
>
> — Laozi, *Tao Te Ching*

In complex real-world operational environments, core power transformers serve not only as the energy dispatch hubs of distribution networks but also as heavy assets with extremely high capital intensity. Conventional distribution asset management often relies on fixed-cycle preventive maintenance, which appears highly inefficient today amidst the surge of massive data in smart grids [13], [14]. Unplanned outages of transformers are usually accompanied by widespread power failure faults and expensive equipment replacement costs. Therefore, how to achieve "optimal utilization" of such core assets and maximize their life-cycle value without altering the existing physical topology of the grid has become a critical topic in modern distribution network management.

To echo the core philosophy of maximizing asset value, the underlying philosophy of the Bianque System is deeply rooted in the concept of "Tao follows nature" from Laozi's *Tao Te Ching*. The mapping of "Tao follows nature" in engineering diagnostics is to conform to the inherent operational ecology of the assets, deeply mining and maximizing the intrinsic value of all existing collected data through algorithms. This study does not intend to add any new hardware sensors, nor does it focus on demonstrating new physical collection methods, but rather focuses entirely on existing conventional data streams. Currently, low-frequency telemetry data represented by Dissolved Gas Analysis (DGA) and Supervisory Control and Data Acquisition (SCADA) systems, as well as high-frequency Digital Fault Recorder (DFR) data that are often only called upon after a fault, have long been used in isolation merely as binary alarm triggers based on static thresholds (e.g., "normal" versus "limit-exceeding") [6]. Such lagging static threshold diagnosis is not only prone to severe false alarms and missed detections but also completely ignores the implicit nonlinear degradation trajectories between multi-dimensional time-series features [15]. We apply "Algorithmic Alchemy" [5] to perform feature extraction and reconstruction on these highly noisy and non-stationary time series, transforming discrete threshold alarms into a quantified countdown of the dynamic Remaining Useful Life (RUL) of the assets, thereby truly achieving the ultimate utilization of data.

Addressing the starkly different physical forms among distribution assets, this paper proposes a dual-track survival auditing architecture for transformers:

1) **Oil-immersed Track (Bionic Telomere Hypothesis):** Based on the physically renewable attribute of insulating oil, the Hayflick Limit [7] is utilized to find a multi-dimensional balance between transformer oil regeneration and the irreversible aging of insulating paper.
2) **Dry-type Track (Degradation Triad Model):** Addressing the pain point of dry-type transformers lacking early chemical early warning from oil, this track physically traces the evolutionary path from chronic thermodynamic fatigue to acute dielectric damage.

Furthermore, this framework transcends pure equipment condition diagnosis by deeply integrating survival analysis with optimal stopping theory, aiming to explore the economic optimal solution for extreme asset sweating through numerical simulation and deduction.

The subsequent structure of this paper is organized as follows: Section II deeply deconstructs the dual-track survival architecture and degradation mathematical models for oil-immersed and dry-type transformers; Section III details the "Six-Mirror" and Bayesian nested mathematical engines responsible for survival baseline optimization and combating dual uncertainties; Sections IV and V present the specific settings of the numerical simulation environment and the sensitivity economic evaluation results based on the Bellman equation, respectively; Section VI introduces the Laplace Platform, which captures extreme transient mutations and provides endless closed-loop feedback for the diagnostic ecology; Section VII is the conclusion of the paper.

II. Dual-Track Survival Architecture

The third part of the Bianque System [2], [3] divides the transformer family into two distinctly different survival auditing tracks, yet they are unified within a single underlying mathematical framework based on the Hazard Function.

A. Track A: Oil-Immersed Transformers and the Bionic Telomere Hypothesis

Oil-immersed transformers possess a unique "renewable" attribute. Building upon the survival auditing core of the first part of core distribution assets (underground cables) [2], we propose the "Bionic Telomere Degradation Hypothesis" to model such assets with local self-recovery characteristics.

In biology, after each cell division, its telomeres experience unidirectional shortening, ultimately leading to apoptosis [7]. We map this mechanism to the thermodynamic and chemical cycles of oil-immersed transformers:

- **Renewable Health Bar (Insulating Oil / DNA):** Transformer oil can recover its dielectric strength through degassing and filtering operations. The system deduces the economic trigger point for the next oil filtration based on the gas generation rate from Dissolved Gas Analysis (DGA).
- **Non-renewable Health Bar (Insulating Paper / Telomere):** The Degree of Polymerization (DP) of the insulating paper suffers irreversible physical breakage after each thermal stress impact. Extensive insulation physics research indicates that the breakage of cellulose chains is a complex chemical kinetic process that directly determines the physical limits of the transformer's resistance to short-circuit electrodynamic shocks [16]. While oil filtration operations can remove moisture and polar impurities to improve the breakdown voltage of the oil body, they cannot reverse this decay in the degree of polymerization [17].

To quantify this physical mechanism, we construct a multi-stage multiplicative decay model based on the underlying logic of the Proportional Hazards Model. Let the basic hazard function of a transformer in a completely natural aging state without any oil filtration intervention follow a two-parameter Weibull Distribution [8], i.e., $\lambda_0(t) = \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1}$. Regarding parameter constraints, the shape parameter must satisfy $\beta > 1$ to indicate that the asset has irreversibly entered the wear-out phase of accelerated aging over time; the scale parameter $\eta > 0$ represents the characteristic life of the equipment.

When the system executes the k-th insulating oil filtration (i.e., "DNA refresh") at time point τ_k, the dielectric performance of the transformer's oil body is partially restored. However, since the decline in the degree of polymerization of the insulating paper (telomeres) is an irreversible structural damage, the overall breakdown resistance capability does not return to its initial state. Therefore, we introduce the "telomere shortening penalty factor" γ_k, defining the effective hazard function $\lambda_k(t)$ after the k-th oil filtration as:

$$\lambda_k(t) = \lambda_{k-1}(t) \cdot \gamma_k \quad (t > \tau_k) \tag{1}$$

In constructing this equation, the system adopts multiplicative logic rather than additive logic (Additive Hazards). Its physical root lies in the fact that the damage caused by the breakage and shedding of solid insulating paper to the vulnerability of the overall system is not linearly additive, but global and geometrically magnified. This conclusion has been widely corroborated in reliability engineering, indicating that multi-source damage exhibits multiplicative coupling rather than additive superposition within insulating materials [18]. Once the insulation structure is damaged, even if a brand-new insulating oil is replaced, the breakdown probability triggered by local electric field distortion will still surge multiplicatively. Therefore, the constraint condition of the multiplier factor is strictly defined as $\gamma_k \in (1, +\infty)$, and it exhibits a monotonically increasing property with the increase of thermal cycles, i.e., $\gamma_{k+1} > \gamma_k$.

Through the above recursive formula, a physical inevitability can be mathematically deduced: due to the continuous amplification of the multiplier γ_k, the decay rate of the transformer's insulation health will exacerbate over time. No matter how frequently the oil filtration operation is executed, the effective safe operational time gained by each intervention (i.e., the time interval between two adjacent oil filtrations $\Delta\tau_k = \tau_k - \tau_{k-1}$) will show a monotonically shrinking trend. To this end, we transform the *Hayflick Limit* in biology into an Optimal Stopping Problem in predictive maintenance [2]. By constructing a Markov Decision Process (MDP) and Bellman's principle of optimality, the system can execute aggressive Asset Sweating strategies while ensuring the grid's safety bottom line, thereby postponing massive capital expenditures [19], [20]. To accurately solve this non-linear evolutionary process in the continuous time domain, this problem is mathematically and rigorously reconstructed as a Partial Integro-Differential Equation (PIDE) with a free boundary. The system seeks the Bellman extremum boundary corresponding to this PIDE through numerical approximation. When the intervention marginal cost C_{filter} of the $(k+1)$-th oil filtration is greater than or equal to the risk-discounted expected revenue gained by this oil filtration, it satisfies:

$$C_{filter} \geq \int_{\tau_k}^{\tau_k+\Delta\tau_{k+1}} r(t) \cdot S_k(t)dt \tag{2}$$

In the above decision boundary, the scalar C_{filter} represents the deterministic marginal costs, such as material consumption and downtime resets, brought about by executing a single insulating oil filtration. During revenue assessment, the system does not simply linearly accumulate the unit time revenue

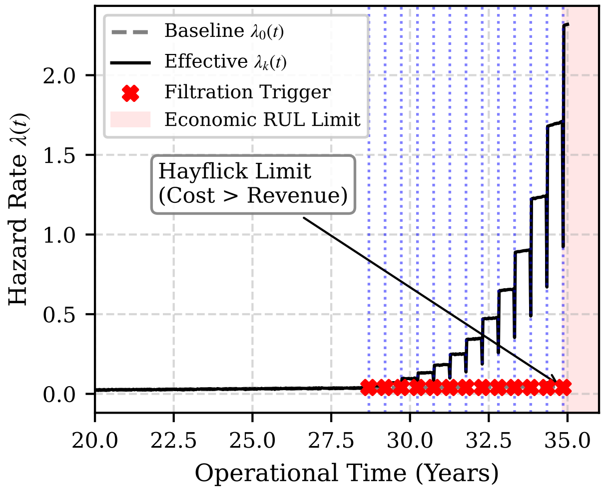

Fig. 1. Evolutionary diagram of the "Bionic Telomere" dual-health-bar degradation and Hayflick Limit for oil-immersed transformers (based on numerical simulation deduction).

$r(t)$ within the next theoretical operational interval $[\tau_k, \tau_k + \Delta\tau_{k+1}]$. This is because, within the extended service cycle, the transformer continuously faces the risk of sudden breakdown that could happen at any time. To scientifically quantify this uncertainty, the algorithm multiplicatively couples the expected revenue $r(t)$ with the survival function $S_k(t) = \exp\left(-\int_0^t \lambda_k(u)du\right)$ (constraint domain $S_k(t) \in [0,1]$) of the corresponding interval. The physical and economic essence of this integral term is to strictly solve for the "Risk-discounted Expected Revenue" while bearing non-stationary decay risks.

To intuitively demonstrate this non-linear decay process, the system generated a numerical simulation deduction chart based on a Monte Carlo Weibull distribution and random normal noise (see Fig. 1). In the evolutionary path, the baseline hazard function $\lambda_0(t)$ (gray dashed line) reflects the irreversible deterioration trajectory of the insulating paper (telomeres). Whenever the DGA gas generation rate reaches the threshold, the system triggers the oil filtration operation (red cross marks and blue vertical dashed line anchors), causing the system to revert to the effective hazard function $\lambda_k(t)$ (black solid line). It can be clearly observed that, with the continuous amplification of the penalty factor γ_k in the later stages of the life cycle, the effective operational time interval $\Delta\tau_k$ gained by each intervention exhibits an obvious accelerated shrinking characteristic. When the marginal cost of high-frequency oil filtration exceeds its extended probabilistic revenue (the red shaded area on the right side of the figure), the Bellman inequality is formally established. This intersection marks the formal touch of the asset to the "Hayflick Limit", which rigorously locks in its final Economic Remaining Useful Life (Economic RUL) in mathematical deduction, and the system will directly trigger the whole-machine replacement procedure.

B. Track B: Dry-Type Transformers and the Degradation Triad

Due to the lack of insulating oil to provide early chemical warnings, dry-type transformers experience irreversible physical and electromagnetic degradation. In this track, the system deeply inherits the Time-Dependent Covariates survival baseline model we verified in the previous cable management [2], and identifies a "Degradation Triad" evolutionary chain based on the physical characteristics of dry-type transformers:

1) **Trigger (THD):** Severe Total Harmonic Distortion leads to a passive increase in the stray losses of the iron core and windings. Literature [21] points out that high-frequency harmonic injection will significantly deteriorate the skin effect and proximity effect of the transformer, becoming a hidden thermodynamic killer.
2) **Chronic Fatigue (ΔT):** Stray losses are converted internally into abnormal dynamic temperature rises under normalized load, plunging the epoxy resin casting body into a vicious cycle of accelerated aging. Epoxy resin will undergo glass transition and microscopic pyrolysis under long-term non-stationary thermal stress [22].
3) **Acute Fault (PD):** Due to long-term thermal stress fatigue, microscopic cracks are generated in the epoxy resin, which in turn trigger irreversible Partial Discharge. Upon entering this stage, the repetition rate and apparent discharge magnitude of the discharge pulses will deviate from linear growth and shift into an extremely dangerous exponential avalanche phase [23].

Mathematically, we embed the aforementioned "Degradation Triad" chain reaction into the classic statistical Time-Dependent Cox Proportional Hazards Model [9]. This model breaks through the static baseline limitations of traditional survival analysis, enabling continuous time integration and risk quantification for dynamically drifting covariates [24]. Given the external harmonic stress time-series feature $X_{THD}(t)$ and the internal thermal fatigue feature $X_{\Delta T}(t)$ observed at continuous time t, the real-time hazard function equation of the system during the chronic fatigue phase of the dry-type transformer is constructed as:

$$\lambda(t|X) = \lambda_0(t) \exp\left(\alpha_1 X_{THD}(t) + \alpha_2 X_{\Delta T}(t)\right) \quad (3)$$

The logical construction of this deduction formula contains two core layers of deconstruction: First, $\lambda_0(t)$ on the right side of the equation is the system's Baseline Hazard Function (constraint domain $\lambda_0(t) > 0$), representing the inherent physical aging rate of the transformer's insulating material over time without any external covariate influence. Second, the system uses the exponential function $\exp(\cdot)$ to wrap the time-dependent covariates and act as a Multiplier on the baseline risk, rather than using a simple Additive Model. The physical and statistical root of this multiplicative combination logic is that the damage caused by electromagnetic harmonics and abnormal temperature rise to epoxy resin is not independently superimposed but plays an exponentially catalytic amplification role. The α_1 and α_2 in the equation are non-negative regression coefficients fitted from numerical simulation samples via Maximum Partial Likelihood Estimation (constraint boundaries $\alpha_1, \alpha_2 \geq 0$), which strictly quantify the sensitivity weights of the transformer insulation to harmonic distortion and temperature rise, respectively.

Through the deconstruction and operation of the above formula, the algorithm only needs to input the time-dependent feature slices of $X_{THD}(t)$ and $X_{\Delta T}(t)$ in the simulation environment to output the dynamically amplified instantaneous physical failure probability $\lambda(t|X)$ at the current time node. Relying on this mechanism, the system achieves a quantitative audit of insulation thermal damage during the lengthy chronic phase. As shown in the numerical simulation deduction process demonstrated in Fig. 2, the system monitors the trigger feature (light blue THD step stress) and its driven fatigue feature (red ΔT dynamic temperature rise) during the long-term chronic phase. Once the internal Partial Discharge (PD) intensity (black solid line) is monitored to erupt exponentially in the multi-dimensional feature space, triggering the vertical orange thick dashed line (Point P) representing the starting point, the macro proportional hazards model based on partial likelihood estimation will no longer be applicable. At this time (the orange shaded area at the bottom of the figure), the algorithm architecture will completely discard the original stationarity assumption and smoothly switch to the acute "Decile Bayesian Funnel" convergence mechanism to combat the extreme survival non-linearity during the system's rush toward the absolute dielectric breakdown limit of the material (Point F of the right red solid line). To ensure an absolute physical defense line while extracting residual value, the system hardcodes an insurmountable "Grade 9" fuse bottom line [3] within the funnel, forcing an automatic power-off directive at the very last moment before a sudden explosive breakdown.

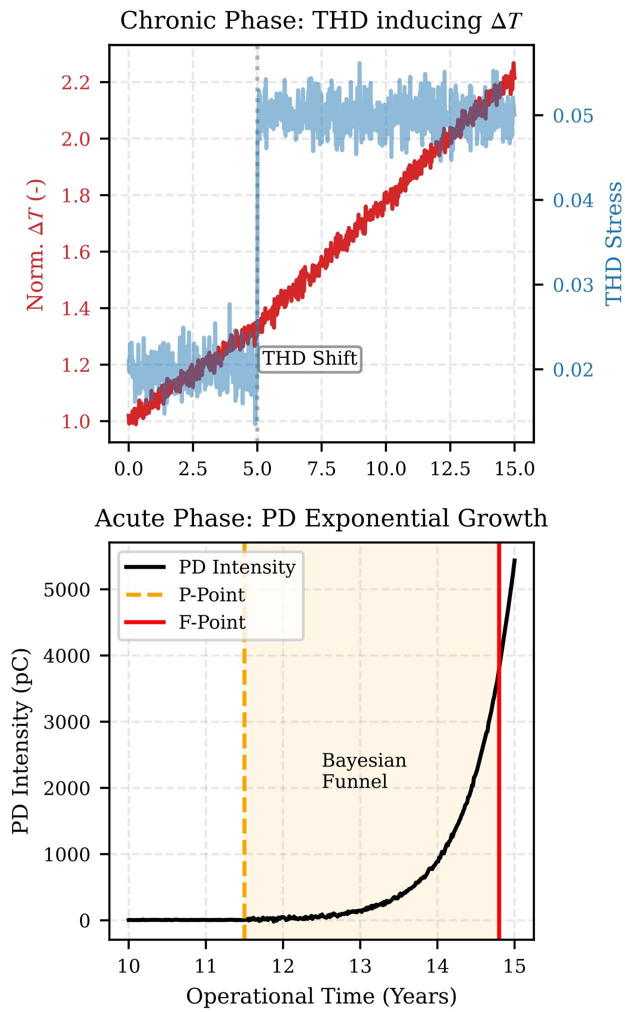

Fig. 2. Evolutionary diagram of the "Degradation Triad" in the time-dependent survival model for dry-type transformers (based on numerical simulation deduction).

III. Core Strategy: Six-Mirror Racing and Bayesian Convergence

The degradation path of transformers under complex physical conditions or underground interwoven environments is filled with high uncertainty. To avoid survival distribution deviations caused by a single statistical model, the system constructs a parallel competitive diagnostic panel called the "Six-Mirror."

A. Six-Mirror Racing Mechanism and Data Defect Compensation

During the routine auditing phase, various time-dependent features extracted through numerical simulation are synchronously input into six independent survival baseline models (covering Weibull, Lognormal, Exponential, Gompertz, Log-logistic, and non-linear proportional hazard variants extended based on neural networks [9]).

The core of the "racing" mechanism is dynamic evaluation based on Maximum Likelihood Estimation (MLE) [1]. Under real complex physical conditions, due to the presence of a large number of operating equipment that have not yet failed, directly applying standard probability models will severely underestimate the overall life of the assets. To this end, when constructing the Likelihood Function ($\mathcal{L}$) of each baseline model, we forcefully introduced a data defect compensation mechanism targeting Right-censoring (indicating equipment that has not failed up to the current audit point). In real industrial equipment ledgers, the vast majority of in-service assets belong to right-censored data, and directly excluding these samples would lead to severely underestimated life predictions [25]. Its underlying mathematical combination logic is shown in the following equation:

$$\mathcal{L}(\theta|\mathcal{D}) = \prod_{i\in\mathcal{F}} f(t_i|\theta) \prod_{j\in\mathcal{C}} S(t_j|\theta) \tag{4}$$

This formula is strictly partitioned into the continuous multiplication of two sets of independent probabilities in its deconstruction. First, the first term on the right side of the equation targets the simulation sample set $\mathcal{F}$ with confirmed faults, and its exact probability of physical failure occurring at specific time t_i is constrained by the probability density function $f(t_i|\theta)$. Secondly, the second term targets the right-censored sample set $\mathcal{C}$ that has not yet experienced a fault. Since their future failure time is unknown, the system no longer forcibly predicts a single breakdown point, but rigorously incorporates their probability of "successfully surviving to the current time t_j," namely the survival function $S(t_j|\theta)$ (constraint domain $S \in [0, 1]$), into the calculation pool.

Here, the physical and statistical essence of the formula adopting the continuous multiplication operator ($\prod$) instead of linear addition lies in: assuming the various transformer assets in the simulation deduction undergo independent physical decay, then given a set of preset parameter matrices θ, the joint probability of the entire sample set presenting this state of "some dead, some still surviving" must be coupled and solved through the product of individual event probabilities. By executing this maximum likelihood equation, the algorithm can calculate the joint probability for each assumed model explaining the current time-series evidence, thereby seamlessly transitioning to the logarithm-based Akaike Information Criterion (AIC) to elect the optimal baseline model with the highest goodness-of-fit. In the field of engineering model selection, AIC has been proven to be one of the most effective information-theoretic criteria balancing the maximum likelihood function and parameter dimensions [26]:

$$AIC_m = 2k_m - 2\ln(\mathcal{L}_m(\hat{\theta}|\mathcal{D})) \tag{5}$$

This criterion was proposed by statistician Hirotugu Akaike, and its core physical significance is to find the ultimate balance between "fitting accuracy" and "model complexity" (i.e., Occam's Razor principle in engineering). In the combinatorial logic of the formula, the second term on the right, $-2\ln(\mathcal{L}_m)$, represents the "fitting bias" of the model: the higher the maximum joint probability $\mathcal{L}_m(\hat{\theta}|\mathcal{D})$ derived based on Formula 4, the smaller the value of this logarithmic penalty term, indicating the model has stronger explanatory power over real ledger fragments. However, blindly pursuing extreme explanatory power can easily lead to adding too many useless parameters to the model, ultimately falling into the trap of "Overfitting" and completely losing the generalization prediction capability for future dynamic survival trajectories. Therefore, the first term on the right side forcefully introduces a model complexity penalty factor $2k_m$ (where k_m is the number of free parameters of the m-th mirror baseline model).

Deduction Results: During each evaluation cycle, the system rigorously calculates and cross-compares the AIC scores of all baseline models in the "Six-Mirror" panel in parallel. Since the algorithm aims to minimize information entropy loss, the model with the lowest score is determined to be the current absolute winner. This winning model is not only used to output the dynamic hazard function for the current cycle, but the system also reversely deduces and locks onto the deep-seated physical deterioration mechanism currently dominant within the transformer insulation system by analyzing its unique probability distribution properties (for instance, if the Weibull distribution wins, it indicates early-to-mid term wear-out; if the Gompertz distribution wins, it indicates highly non-linear extreme collapse near the end of life).

B. Bayesian Nested Simulation to Combat Dual Uncertainties

Once the transformer approaches the terminal interval of the life distribution (e.g., entering Point P of partial discharge for dry-type transformers, or the high-frequency oil filtration cycle stage for oil-immersed transformers), the system abandons coarse-grained stationary predictions and forcefully initiates the Bayesian Convergence mechanism to combat extreme survival non-linearities.

To handle the "dual uncertainties" faced by the system, we established a two-layer Bayesian Nested Monte Carlo Simulation architecture:

- **Outer Loop (Parameter Convergence for Epistemic Uncertainty):** Targeting the winning parameterized model from the "Six-Mirror," the system sets its output coarse parameter distribution as the prior probability density $P(\theta)$ (constraint range: $\theta \in \Theta$). Whenever the system captures new DGA or SCADA time-dependent feature fragments from complex physical conditions (defined as new evidence $\mathcal{D}_{new}$), the outer engine reshapes the posterior distribution of the parameters based on the classic Bayes' Theorem:

$$P(\theta|\mathcal{D}_{new}) = \frac{P(\mathcal{D}_{new}|\theta)P(\theta)}{\int_{\Theta} P(\mathcal{D}_{new}|\theta)P(\theta)d\theta} \tag{6}$$

  The deconstruction of this formula is as follows: this is a typical probability multiplicative coupling and normalization process. In the numerator, the prior probability $P(\theta)$ represents the system's "initial belief" in the asset's life before seeing new data; the likelihood function $P(\mathcal{D}_{new}|\theta)$ quantifies the absolute probability of this new evidence occurring under this parameter belief. Multiplying the two is essentially using real-world evidence to penalize or reward the initial belief. The denominator (marginal likelihood) is the probability integral over the entire parameter space Θ, and its purely mathematical role is normalization constraint, ensuring the calculated posterior distribution $P(\theta|\mathcal{D}_{new})$ strictly satisfies probability axioms in its value domain (i.e., the total integral is 1). Through this iterative deduction based on Bayesian Bootstrapping, the system overcomes the epistemic blind spots of static parameters, achieving a comprehensive decoupling and quantitative auditing from Epistemic Uncertainty to Aleatoric Uncertainty [29]. Through high-dimensional Markov Chain Monte Carlo (MCMC) deduction or Bayesian bootstrapping, we can break the reliance on parameter point estimates [27], [28]. The specific derived result is: the distribution variance of core model parameters (such as the Weibull scale η) shrinks geometrically as data flows in.
- **Inner Loop (Physical Sampling for Aleatoric Uncertainty):** After the outer layer obtains the continuously converging posterior parameter matrix θ, the inner algorithm executes massive Time-to-Failure (TTF) random sampling based on this posterior distribution, to numerically simulate unpredictable physical emergencies such as discharge avalanches or thermodynamic breakdown.

The final output of this grand nested deduction process is not a cold scalar, but a highly focused "Bayesian Funnel" of survival probability density as shown in Fig. 3. The underlying data generation mechanism of this figure is based on a normal kernel function whose parameters gradually shrink,

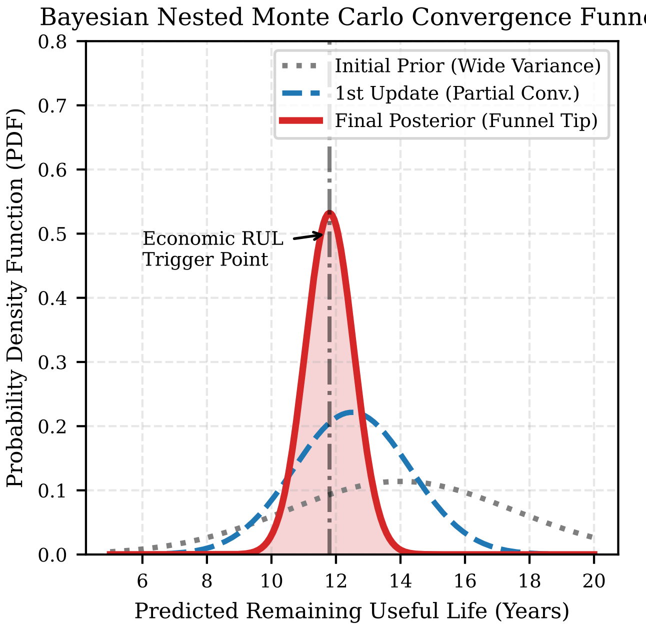

Fig. 3. Probability density convergence funnel in Bayesian nested Monte Carlo simulation deduction. The distribution curves demonstrate the dynamic collapse process where the variance of survival prediction shrinks geometrically after the system absorbs time-series evidence.

superimposed with Monte Carlo random sampling simulating high-frequency DGA sampling cycles for numerical simulation deduction.

The legend analysis of this funnel is as follows: The horizontal axis in the figure represents the Remaining Useful Life (RUL) predicted by the model, and the vertical axis represents the corresponding probability density at that life point. The gray dotted line represents the high-variance, broad parameter prior distribution when the outer loop just starts, due to a lack of time-series evidence. As the deduction proceeds, the blue short dashed line represents the initial gathering of the probability distribution after the system absorbs the mid-term SCADA covariate evidence. Ultimately, the red thick solid line and its underlying light red shaded area represent the posterior probability density distribution with extremely low variance formed by complete parameter collapse after passing through the double-layer nested loops. The deduction result is very clear: the tip of the red funnel accurately pointed to by the black long dashed line and black arrow is the Economic RUL Trigger Point that the system ultimately locks onto amidst extreme uncertainty. Combined with the optimal stopping theory described in Formula 2, the algorithm engine directly aligns this tip with the "Grade 9" safety bottom line [3], and takes it as the integration boundary of the penalty factor, thereby maximizing the extraction of the transformer asset's economic residual value while ensuring an absolute physical defense line.

IV. Numerical Simulation Environment and Algorithm Flow

According to the highest declaration of academic integrity principles, **the data foundation used throughout this paper (including all charts used to display theoretical evolutionary paths in preceding sections, and subsequent specific analysis results)** does not originate from on-site measurements of any specific physical power grid, but is entirely based on a Numerical Simulation built upon strict physical mechanisms and statistical probability logic. This method aims to detach from the narrow physical constraints of specific regions or specific equipment, purely verifying the universality and theoretical limits of the dual-track survival architecture on a mathematical and logical level.

A. Simulation Parameter Settings

The underlying simulation of the system is established based on multi-level Monte Carlo sampling, with environment settings as follows: First, in "Track A (Oil-Immersed)," $N = 10,000$ virtual transformer entities are initialized for Monte Carlo sampling. Their insulating paper life baseline follows a Weibull distribution, with scale parameter set to $\eta = 40$ years and shape parameter $\beta = 2.2$. The introduced telomere shortening penalty factor γ_k follows a geometric amplification logic, forcefully amplified after each oil filtration intervention, i.e., the single incremental penalty multiplier is set to 1.35. Second, in "Track B (Dry-Type)," the generation of the time-dependent covariate $X_{THD}(t)$ employs a step function with Brownian Motion noise to simulate random impacts of grid harmonics; the partial discharge $X_{PD}(t)$ signal is set to switch into exponential burst mode once it crosses a certain cumulative threshold.

B. Bayesian Nested Deduction Pseudocode

To realize the deduction combating dual uncertainties in the core strategy, the system executes the following algorithm flow:

V. Simulation Results Analysis and Economic Evaluation

A. Degradation Chain and Survival Probability Funnel Deconstruction

Based on the simulation architecture above, the output results of the system possess clear physical and statistical interpretability. First, observe the degradation triad evolutionary chart of Track B (Fig. 2). In the chronic fatigue phase in the upper half, the simulation system clearly records that when the service time is $t = 5$ years, the THD (Total Harmonic Distortion) time series undergoes a step surge due to external grid pollution. After a brief lag of physical heat accumulation, this trigger directly drives the steep deflection of the slope of the ΔT feature, quantifying the immediate catalytic effect of the time-dependent covariate on the baseline risk $\lambda_0(t)$. With the accumulation of chronic damage, in the acute phase in the lower half, when running to $t = 11.5$ years (i.e., Point P), the Partial Discharge (PD) signal breaks through the threshold and

Algorithm 1 Transformer Economic Life Auditing Based on Bayesian Nested Monte Carlo

Require: Historical monitoring time-series data $\mathcal{D}_{past}$, incoming observation data $\mathcal{D}_{new}$, winning baseline model $\lambda_0(t)$ from Six-Mirror
Ensure: Posterior parameter matrix θ_{post}, Economic RUL decision point T_{RUL}
Initialize Outer Epistemic Deduction:
Calculate prior probability density $P(\theta_{prior})$ via MLE based on $\mathcal{D}_{past}$
while Receiving new high-frequency monitoring features $\mathcal{D}_{new}$ **do**
  Step 1: Outer Bayesian Parameter Update
  Calculate likelihood $\mathcal{L} = P(\mathcal{D}_{new}|\theta_{prior})$
  $P(\theta_{post}) \leftarrow \frac{\mathcal{L}\cdot P(\theta_{prior})}{\int \mathcal{L}\cdot P(\theta)d\theta}$
  Extract variance-shrunk posterior parameter distribution matrix θ_{post}
  Step 2: Inner Physical Random Sampling
  for $i = 1$ **to** $N_{samples}$ **do**
    Draw specific parameter set θ_i from $P(\theta_{post})$
    Combine with time-dependent covariate $X(t)$, simulate single TTF_i
  end for
  Step 3: Construct Survival Probability Funnel
  Fit probability density curve $f(t)$ for $TTF_{1 \ldots N_{samples}}$
  Step 4: Bellman & PIDE Optimization
  if $\int_t^{t+\Delta t} r(u)S(u)du \leq C_{filter}$ **then**
    $T_{RUL} \leftarrow t$
    Break and output final economic replacement directive
  end if
end while

turns into an exponential avalanche. At this time, the system algorithm discards the stationarity assumption and initiates the evolutionary deduction towards the dielectric breakdown point (Point F, predicted to be 14.8 years).

At the macroscopic level of uncertainty convergence, Fig. 3 demonstrates the mechanism of Bayesian nested deduction. In the initial stage, the prior distribution (gray dotted line) presents a high-variance flat shape with a mean at 14.0 years, representing the system's high degree of epistemic uncertainty at this time. As the deduction proceeds, after absorbing continuous mid-term SCADA covariate evidence, the first round of posterior distribution (blue short dashed line) gathers to the left to 12.5 years. Ultimately, the tip of the red low-variance funnel is fixed at 11.8 years, verifying the Bayesian theoretical expectation that "the closer to the physical boundary of failure, the more the prediction variance shrinks."

B. Sensitivity Analysis under the Hayflick Limit

The core argument of this section is: **In modern capital-intensive distribution networks, the "Remaining Useful Life (RUL)" of heavy assets is not a pure physical constant, but an economic boundary constrained by the dynamic game between maintenance costs and operational revenue.** To prove that this framework possesses the attributes of a financial decision engine (CFO), we conducted sensitivity deductions on the core economic parameter C_{filter} (comprehensive marginal cost of a single oil filtration).

In the default simulation baseline, assume C_{filter} is set to 1.0 unit, while the base operational revenue $r(t)$ per unit time remains constant. With the cumulative amplification of the telomere penalty factor γ_k, the transformer insulating paper becomes increasingly fragile, and the safe operational time gained per oil filtration becomes shorter and shorter. When the system deduces that it is ready to perform the $k = 4$-th oil filtration, due to the severe shrinkage of the effective safe operational interval, the risk-discounted expected revenue within this interval falls below 1.0 unit (i.e., integral revenue $\leq C_{filter}$). At this point, the Bellman inequality of Formula (2) holds, the system determines the asset has touched the "Hayflick Limit," issues a financial directive to replace the transformer, and refuses to execute the unprofitable 4th oil filtration.

To verify the acuity of this decision model, we introduced external economic turmoil variables into parallel simulations. Assume that due to supply chain shortages or rising labor costs, the single oil filtration cost C_{filter} increased by 20% (becoming 1.2 units). It should be noted that in this scenario, the physical degradation trajectory within the transformer ($\lambda_0(t)$ and γ_k) remains unchanged. However, due to the substantial rise in intervention costs, the intersection where revenue drops below costs is brought forward. The algorithm engine automatically responds, advancing the trigger point of the Hayflick Limit from the $k = 4$-th to the $k = 3$-rd oil filtration cycle.

This sensitivity deduction yields a conclusion of clear guiding significance: **Under identical physical deterioration states, assets with higher maintenance costs will arrive at their "economic replacement point" earlier.** By capturing fluctuations in economic variables, the system altered the static engineering logic of deciding maintenance relying purely on physical states, verifying the dynamic financial gaming mechanism during the Asset Sweating process achieved through pure data algorithms.

VI. Laplace Ecological Synergy: Microsecond Autopsy and Endless Loops

A. The End of Agnosticism: Capturing Black Swan Faults

In a survival auditing system built on statistical prior distributions, extreme "Black Swan" events [10] are always critical factors affecting prediction confidence. No matter how rigorous the evolutionary path of the Bianque System's "Six-Mirror" is, it essentially remains a low-frequency monitoring architecture handling "chronic progressive decay" (such as month-level insulating paper aging, or day-level partial discharge accumulation). Once the transformer encounters devastating external impacts (such as microsecond-level lightning intrusion waves, millisecond-level external short-circuit electrodynamic shocks) or instantaneous collapse of latent microscopic manufacturing defects, the conventional low-frequency SCADA system has no time to react before the asset fails.

For the existing operational maintenance system, such sudden failures mean prediction failure. However, in the underlying logic of the Bianque System, every "unpredicted death" that causes model collapse is nourishment driving diagnostic ecological evolution. This is the foundational meaning of the existence of its homologous system—the *Laplace Platform*.

B. Holographic Time-Scale Retrospection and Algorithmic Autopsy

When a transformer experiences a sudden failure, the first action of the Laplace Platform is not limited to the mere instant before death, but, akin to an aviation "Black Box", initiates a "long-term + short-term" multi-tier holographic retrospection. The system will extrapolate the timeline backward, sequentially extracting time-dependent feature slices of the equipment 1 week, 1 day, 1 hour, 1 minute, and down to the final 30 milliseconds before death, forming a "Failure Holographic Memo" for post-event analysis.

Within this cross-scale time funnel, the Laplace Platform performs a multi-dimensional "algorithmic autopsy." To concretize this deduction process, Fig. 4 displays a slice matrix constructed by the system based on numerical simulation. Its underlying data is generated by a steady-state fundamental wave containing Gaussian white noise superimposed with a high-frequency decay function. The chart is divided into three tiers from top to bottom, respectively corresponding to marks and physical meanings of different colors: First, the Top Panel projects the macroscopic SCADA low-frequency telemetry profile for 1 week ($\approx$ 168 hours) prior to the failure. The Solid Gray Line in the figure characterizes the steady-state fluctuation trajectory of oil temperature driven by the grid baseline load. At this scale, the system captures no abnormal drifts exceeding traditional thermodynamic thresholds, which inversely corroborates that this failure did not originate from classical chronic aging of insulating paper or long-term thermal stress overload. Therefore, this long-term background data is utilized by the system for Baseline Alignment to strictly rule out progressive degradation artifacts; meanwhile, the red vertical shaded band on the right side of the figure constitutes a time focusing anchor, directing the algorithmic engine to forcefully converge the horizon towards the meso-scale.

Secondly, the Middle Panel intercepts the highly dynamic dielectric state evolution window for 1 day (24 hours) before the equipment's death. Within this interval, the apparent discharge magnitude of Partial Discharge (PD) characterized by the Solid Blue Line deviates from the conventional stationary Markov evolutionary trajectory, presenting an exponential nonlinear steep surge characteristic rapidly crossing from latency to Avalanche Breakdown. Such a dramatic change in dielectric properties indicates that the interior of the insulating medium is undergoing irreversible microscopic void tearing, but since this process is extremely brief and usually accompanied by severe field broadband noise masking, conventional low-frequency inspections are highly prone to miss detection. Similarly, the dark red shaded block on the right side of this panel acts as a secondary trigger anchor, indicating that the system is rapidly contracting at a micro-scale towards the final moment that decides life and death.

Finally, the Bottom Panel reproduces with high fidelity the extreme transient process of the final 30 milliseconds when the lethal blow occurred. The data capture of this microsecond Digital Fault Recorder (DFR) slice deeply relies on the underlying hardware's Ring Buffer and high-speed Direct Memory Access (DMA) mechanisms [11]. Currently, the most advanced high-speed Analog-to-Digital Converter (ADC) front-end architectures in the industry fully possess the physical conditions to support megahertz (MHz) level transient freezing [32]. In terms of operational logic, the transformer edge gateway does not start recording in advance due to "foreseeing" the fault, but performs normalized dead-zone-free sampling based on the principle of rolling erasure; only at the exact moment when dielectric breakdown occurs and the protection device triggers a trip and power loss, does the system utilize a Post-fault Event Trigger to deadlock and extract this final "death footage." As shown in the figure, within the steady-state profile of the black 50 Hz fundamental voltage, the algorithmic engine injects high-frequency Continuous Wavelet Transform (CWT) for joint time-frequency reconstruction, supplemented by Fractal Dimension analysis, to successfully and precisely strip out the sudden high-frequency damped oscillation cluster pointed to by the Red Arrow on the right side of the figure, enveloped by wide-area broadband electromagnetic background noise. This signal cluster highly likely corresponds to the refracted and reflected impact of a lightning intrusion wave or the instantaneous flashover of inter-turn winding insulation. CWT, with its adaptive scaling and translation window, provides an unparalleled capability to capture transient singularities [30], while the fractal dimension provides an absolute quantitative calibration of this waveform's transition from a steady state to a disordered state from the perspective of chaotic dynamics [31].

This joint slicing mechanism from weeks to milliseconds aims to accurately characterize the sudden mutation trajectory of extreme black swan faults in the ultra-short time before an outbreak.

C. Deriving Something from Nothing: Data Flywheel and Ecological Immunity Remodeling

Once this extremely weak precursor feature is successfully "purified" and mathematically corroborated by the Laplace Platform, it completes a dimensional mapping from "agnostic chaos" to "clear physical parameters." Following this, the Laplace Platform encapsulates this brand new feature into an independent Time-Dependent Covariate and forcefully injects it back into the Bianque System's "Six-Mirror" baseline model repository.

This closed-loop feedback across time scales [12] completely remodels the feature set matrix (Feature Space) of the entire distribution network transformer population. It is just like a biological immune system extracting an antigen and distributing it to all cells after encountering an unknown mutant virus. This Artificial Immune System (AIS) paradigm [33] endows the diagnostic ecology with unprecedented adaptive robustness. When any transformer in normal operation

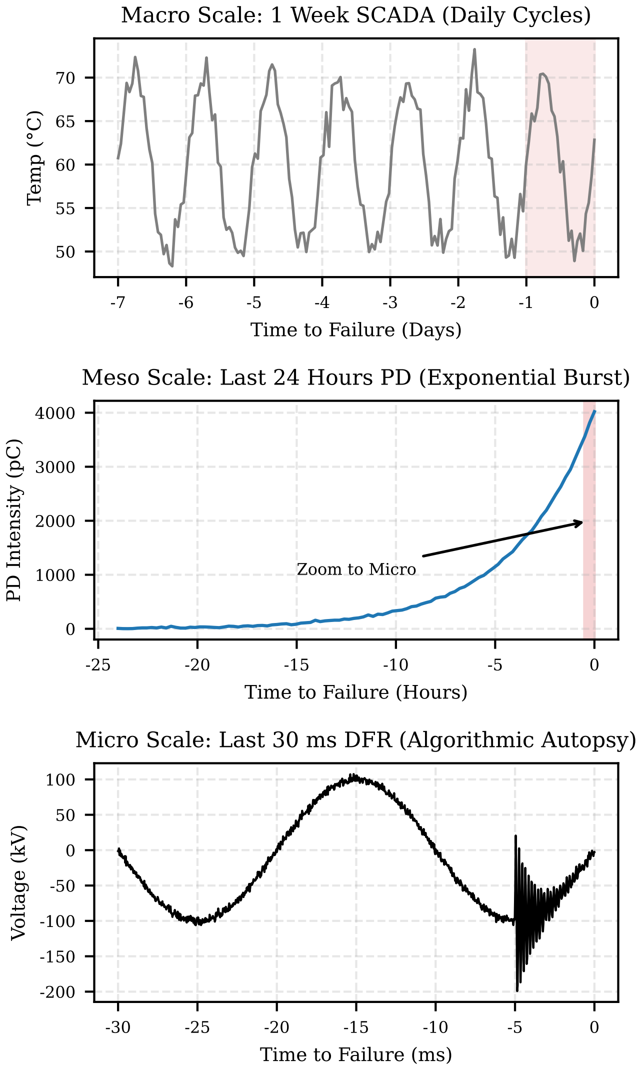

Fig. 4. Cross-scale feature slice diagram of the Laplace Platform's "Holographic Time-Scale Memo" (based on numerical simulation).

across the entire grid flickers with this weak high-frequency feature again on a future day, the Bayesian nested engine of the Bianque System will no longer turn a blind eye, but will trigger an early warning instantaneously.

Through this data flywheel mechanism of "using extreme microscopic transients (Laplace) to feed back into macroscopic slow survival (Bianque)", the destructive casualties of equipment are no longer a shame to the system, but are elevated into a stepping stone pushing the predictive ecology towards adaptive evolution. This is exactly the ultimate engineering mapping of "All things come into being, and I see thereby their return" from Laozi's *Tao Te Ching*—in the endless chaos and death of the physical world, the system achieves infinite convergence and immortality of predictive capabilities.

VII. Conclusion

The third part of the Bianque System breaks through the limitations of existing transformer operational maintenance that overly relies on physical perception and static thresholds, providing a framework integrating bionics philosophy and hardcore statistical algorithms for the management of core heavy distribution grid assets [4].

First, through the "Dual-Track Survival Architecture", the system not only quantifies the financial boundary of oil-immersed transformers under the "Hayflick Limit," but also characterizes the degradation avalanche trajectory of dry-type transformers under non-stationary thermodynamic shocks. Secondly, relying on "Six-Mirror Racing" and "Bayesian Nested Deduction", the system achieves mathematical decoupling of dual cognitive and aleatoric uncertainties, making highly complex failure probabilities converge into intuitive economic countdowns. Finally, as the ultimate defense line of this architecture, the Laplace Platform utilizes hardware-level high-speed caching and microsecond time-frequency analysis to transform destructive black swan faults into "antigen features" feeding back into the system, completely closing the loop of the asset management data flywheel ecology.

In summary, this research provides a predictive method for the operational limits of transformer assets, capable of transforming them into considerable capital replacement dividends through mathematical Asset Sweating, laying a theoretical foundation for the depth defense and investment optimization of future smart grids.

Acknowledgment

In the process of exploring underlying survival algorithms for predictive maintenance of distribution grid assets, this study was profoundly inspired by the research series published by M. R. Shadi et al. [1], [9]. Their outstanding review article [1] provided a comprehensive, systematic, and in-depth retrospective on multi-dimensional survival models in sensor-enabled smart energy networks. Through detailed deconstruction and comparison of fully parametric models, semi-parametric models, and machine learning/deep learning-based survival analysis architectures, the original author team provided academia with a valuable algorithmic panorama and theoretical summary. Furthermore, their recent work on Bayesian nested Monte Carlo simulation under data deficiencies [9] provided crucial methodological insights for handling dual uncertainties in our system.

It is precisely thanks to the high-level perspective and meticulous classification guidance of this literature that this study was able to smoothly establish the "Six-Mirror Panel" auditing mechanism and Bayesian convergence of the Bianque System, accurately completing the selection of the underlying core algorithm library. Herein, we express our most sincere respect and gratitude to the original author team represented by M. R. Shadi, L. K. Mortensen, H. Mirshekali, and H. R. Shaker. Their high-level research achievements laid a solid theoretical cornerstone for this paper.

References

[1] M. R. Shadi, H. Mirshekali, M. Tahavori, and H. R. Shaker, "Survival Models for Predictive Maintenance and Remaining Useful Life in Sensor-Enabled Smart Energy Networks: A Review," *Sensors*, vol. 26, no. 6, p. 1915, 2026. [Online]. Available: https://doi.org/10.3390/s26061915

[2] Y. Zeng, “From Causal Auditing to Value Maximization for Core Distribution Assets—Part I: Underground Cable Management (v1.0.0),” *Zenodo*, 2026. [Online]. Available: https://doi.org/10.5281/zenodo.20676445

[3] Y. Zeng, “From Causal Auditing to Value Maximization for Core Distribution Assets—Part II: Switchgear Management (v1.0.0),” *Zenodo*, 2026. [Online]. Available: https://doi.org/10.5281/zenodo.20751877

[4] Y. Zeng, “From Causal Auditing to Value Maximization: Individualized Management Guidelines for Offshore Wind Assets (v1.0.0),” *Zenodo*, 2026. [Online]. Available: https://doi.org/10.5281/zenodo.20391481

[5] Y. Zeng, “Mitigating Large Language Model Hallucinations in Industrial Environments via Asymmetric Deterministic Override Architecture,” *Zenodo*, 2026. [Online]. Available: https://doi.org/10.5281/zenodo.20539607

[6] A. A. Adekunle, I. Fofana, P. Picher, et al., “Multiclass fault diagnosis in power transformers using dissolved gas analysis and grid search-optimized machine learning,” *Energies*, vol. 18, no. 13, p. 3535, 2025. [Online]. Available: https://doi.org/10.3390/en18133535

[7] A. Bernadotte, V. M. Mikhelson, and I. M. Spivak, “Markers of cellular senescence. Telomere shortening as a marker of cellular senescence,” *Aging (Albany NY)*, vol. 8, no. 1, pp. 3–11, Jan. 2016. [Online]. Available: https://doi.org/10.18632/aging.100871

[8] H. Hirose, “Maximum likelihood parameter estimation in the extended Weibull distribution and its applications to breakdown voltage estimation,” *IEEE Transactions on Dielectrics and Electrical Insulation*, vol. 9, no. 4, pp. 524–536, Aug. 2002. [Online]. Available: https://doi.org/10.1109/TDEI.2002.1024429

[9] M. R. Shadi, L. K. Mortensen, H. Mirshekali, and H. R. Shaker, “Proactive Cable Replacement Planning Using Neural Weibull Proportional Hazard Modeling and Bayesian Nested Monte Carlo Simulation Under Data Deficiencies,” *IEEE Access*, vol. 13, pp. 167823–167836, 2025. [Online]. Available: https://doi.org/10.1109/ACCESS.2025.3613832

[10] A. Gholami, T. Shekari, M. H. Amirioun, *et al.*, “Toward a consensus on the definition and taxonomy of power system resilience,” *IEEE Access*, vol. 6, pp. 32035–32053, 2018. [Online]. Available: https://doi.org/10.1109/ACCESS.2018.2845378

[11] R. P. Medeiros, F. B. Costa, K. M. Silva, *et al.*, “A Clarke-Wavelet-Based Time-Domain Power Transformer Differential Protection,” *IEEE Transactions on Power Delivery*, vol. 37, no. 1, pp. 317–328, Feb. 2022. [Online]. Available: https://doi.org/10.1109/TPWRD.2021.3059732

[12] G. Falekas and A. Karlis, “Digital twin in electrical machine control and predictive maintenance: State-of-the-art and future prospects,” *Energies*, vol. 14, no. 18, p. 5933, 2021. [Online]. Available: https://doi.org/10.3390/en14185933

[13] S. Rana, “AI-driven fault detection and predictive maintenance in electrical power systems: A systematic review of data-driven approaches, digital twins, and self-healing grids,” *American Journal of Advanced Technology and Engineering Solutions*, vol. 1, no. 1, pp. 258–289, 2025. [Online]. Available: https://doi.org/10.63125/4p25x993

[14] J. I. Janjua, M. Nadeem, and Z. A. Khan, “Machine Learning Based Prognostics Techniques for Power Equipment: Comparative Study,” in *2021 IEEE International Conference on Computing (ICOCO)*, Kuala Lumpur, Malaysia, 2021, pp. 265–270. [Online]. Available: https://doi.org/10.1109/ICOCO53166.2021.9673564

[15] H. Shang, Y. Wei, and S. Zhang, “Transformer Fault Diagnosis Based on Knowledge Distillation and Residual Convolutional Neural Networks,” *Entropy*, vol. 27, no. 7, p. 669, 2025. [Online]. Available: https://doi.org/10.3390/e27070669

[16] C. Fernández-Diego, A. Ortiz, I. A. Carrascal, *et al.*, “Kinetic Assessment of Mechanical Properties of a Cellulose Board Aged in Mineral Oil and Synthetic Ester,” *Polymers*, vol. 13, no. 23, p. 4150, 2021. [Online]. Available: https://doi.org/10.3390/polym13234150

[17] D. Sarathkumar, M. Srinivasan, and R. A. Raj, “An extensive critique on transformer oil reclamation techniques,” in *2022 IEEE 7th International Conference on Recent Advances and Innovations in Engineering (ICRAIE)*, Mangalore, India, 2022, pp. 438–443. [Online]. Available: https://doi.org/10.1109/ICRAIE56454.2022.10054184

[18] N. Gorjian, L. Ma, M. Mittinty, *et al.*, “A review on reliability models with covariates,” in *Engineering Asset Lifecycle Management*. London: Springer, 2010, pp. 385–397. [Online]. Available: https://doi.org/10.1007/978-0-85729-320-6_43

[19] D. Wu, K. Gao, R. Peng, *et al.*, “Optimal preventive maintenance policy considering technical-innovation-empowered renewals,” *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability*, 2025.

[20] S. Yi, C. Watts, A. Cooper, *et al.*, “A regulatory approach to the assessment of asset replacement in electricity networks,” in *IET and IAM Asset Management Conference 2011*, Stevenage, UK: IET, 2011, p. 33. [Online]. Available: https://doi.org/10.1049/cp.2011.0550

[21] A. Hamadi, S. Afshar, M. S. Alam, and S. A. Arefifar, “Power Transformer Harmonic and Electro-Thermal Modeling: A Multiphysics Review,” *IEEE Access*, vol. 14, pp. 58590–58618, 2026. [Online]. Available: https://doi.org/10.1109/ACCESS.2026.3683994

[22] X. Kong, C. Zhang, H. S. Du, *et al.*, “Effects of Thermo-Oxidative Aging on the Dielectric Property and Electrical Breakdown of Epoxy Resin Using in High-Voltage Equipment,” *IEEE Transactions on Dielectrics and Electrical Insulation*, vol. 32, no. 1, pp. 254–262, Feb. 2025. [Online]. Available: https://doi.org/10.1109/TDEI.2024.3501997

[23] G. C. Montanari, “Bringing an insulation to failure: the role of space charge,” *IEEE Transactions on Dielectrics and Electrical Insulation*, vol. 18, no. 2, pp. 339–364, April 2011. [Online]. Available: https://doi.org/10.1109/TDEI.2011.5739438

[24] A. Barabadi, J. Barabady, and T. Markeset, “Maintainability analysis considering time-dependent and time-independent covariates,” *Reliability Engineering & System Safety*, vol. 96, no. 1, pp. 210–217, 2011. [Online]. Available: https://doi.org/10.1016/j.ress.2010.08.007

[25] S. Maximov, V. H. Coria, F. Rivas-Davalos, *et al.*, “New analytical method for estimating mean life of electric power equipment based on complete and right-censored failure data,” *Electric Power Systems Research*, vol. 154, pp. 311–318, 2018. [Online]. Available: https://doi.org/10.1016/j.epsr.2017.08.042

[26] H. Akaike, “A new look at the statistical model identification,” *IEEE Transactions on Automatic Control*, vol. 19, no. 6, pp. 716–723, Dec. 1974. [Online]. Available: https://doi.org/10.1109/TAC.1974.1100705

[27] N. Z. Gebraeel, M. A. Lawley, R. Li, and J. K. Ryan, “Residual-life distributions from component degradation signals: A Bayesian approach,” *IIE Transactions*, vol. 37, no. 6, pp. 543–557, 2005. [Online]. Available: https://doi.org/10.1080/07408170590929018

[28] D. B. Rubin, “The Bayesian bootstrap,” *The Annals of Statistics*, vol. 9, no. 1, pp. 130–134, 1981. [Online]. Available: http://www.jstor.org/stable/2240875

[29] I. Ramirez, J. Alcibar, J. Pino, *et al.*, “Disentangling Aleatoric and Epistemic Uncertainty in Physics-Informed Neural Networks: Application to Insulation Material Degradation Prognostics,” *Advanced Intelligent Systems*, p. e202501502, 2026. [Online]. Available: https://doi.org/10.1002/aisy.202501502

[30] D. C. Robertson, O. I. Camps, J. S. Mayer, and W. B. Gish, “Wavelets and electromagnetic power system transients,” *IEEE Transactions on Power Delivery*, vol. 11, no. 2, pp. 1050–1058, April 1996. [Online]. Available: https://doi.org/10.1109/61.489367

[31] S. Haykin and X. B. Li, “Detection of signals in chaos,” *Proceedings of the IEEE*, vol. 83, no. 1, pp. 95–122, Jan. 1995. [Online]. Available: https://doi.org/10.1109/5.362751

[32] Y. Cui, K. A. Lan, and E. V. Hungerford, “A front-end DAQ with buffer and flexible triggers,” *IEEE Transactions on Nuclear Science*, vol. 53, no. 3, pp. 1356–1362, June 2006. [Online]. Available: https://doi.org/10.1109/TNS.2006.873192

[33] Y. Zhang, L. Wang, W. Sun, R. C. Green, and M. Alam, “Artificial immune system based intrusion detection in a distributed hierarchical network architecture of smart grid,” in *2011 IEEE Power and Energy Society General Meeting*, Detroit, MI, USA, 2011, pp. 1–8. [Online]. Available: https://doi.org/10.1109/PES.2011.6039697